MVP Dagen

Foundry Local: Bygg og kjør AI lokalt med Microsoft Foundry



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Agenda

- What is Foundry Local
- Why using Foundry Local
- Foundry Local Architecture
- How to run your own models in Foundry Local
- Hybrid approach
- Q&A

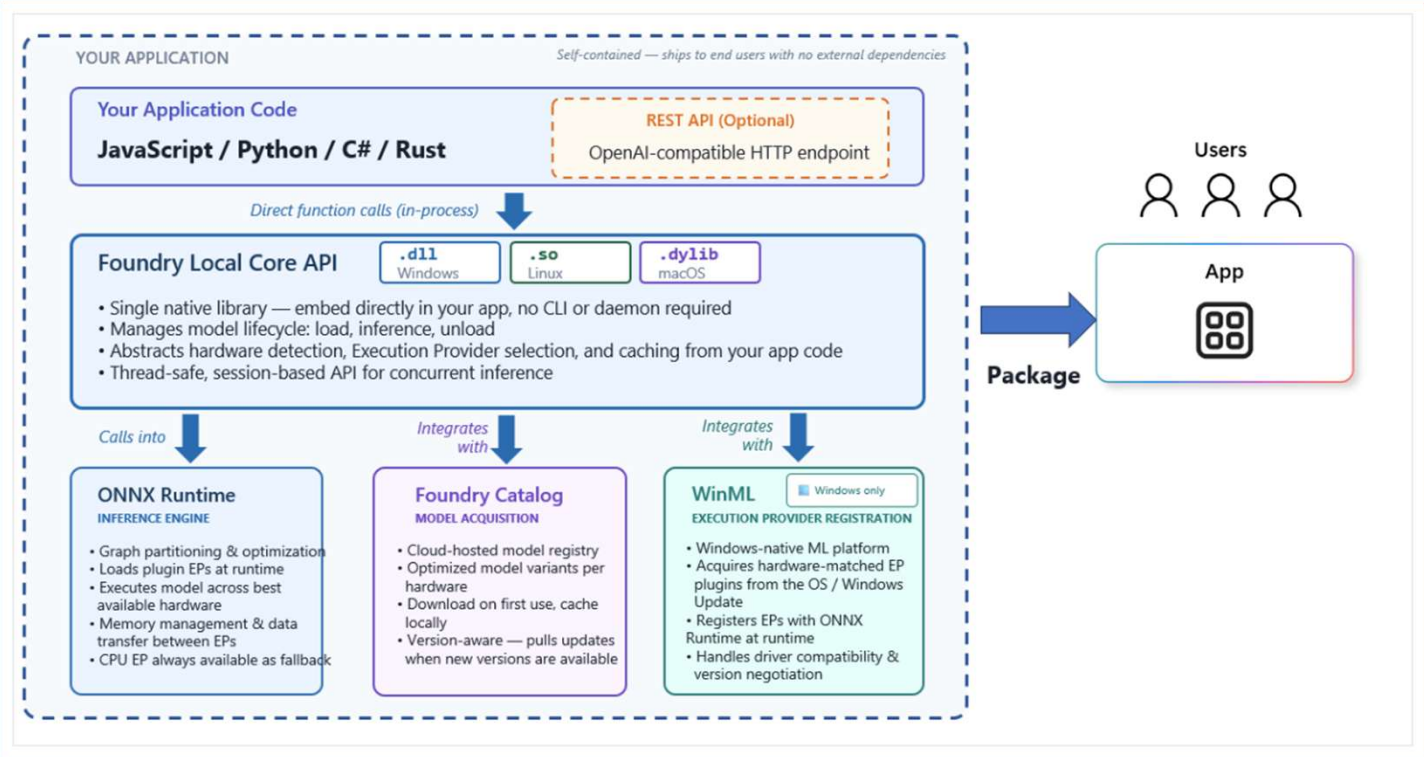
What is Foundry Local?

- Runs AI models directly on the user's device
- SDK for C#, Python, JavaScript and Rust
- Use ONNX runtime as inference engine
- Curated Foundry catalog with optimized model variants

Why using Foundry Local?

- Keep sensitive data on your device
- Operate on limited connectivity or offline environments
- Reduce per-token cloud inference costs
- Deliver low-latency AI response for real time interactions

Foundry Local Architecture



Src: https://learn.microsoft.com/en-us/azure/foundry-local/concepts/foundry-local-architecture

Execution Providers

Execution provider	Device type	Platform
NVIDIA CUDA	GPU	Windows, Linux
WebGPU (via Dawn)	GPU	Windows, Linux, macOS
AMD Vitis	NPU	Windows
Qualcomm	NPU	Windows
Intel OpenVino	GPU	Windows
CPU	CPU	Windows, Linux, macOS

Model lifecycle

- Download model
- Load model
- Inference
- Unload

DEMOS

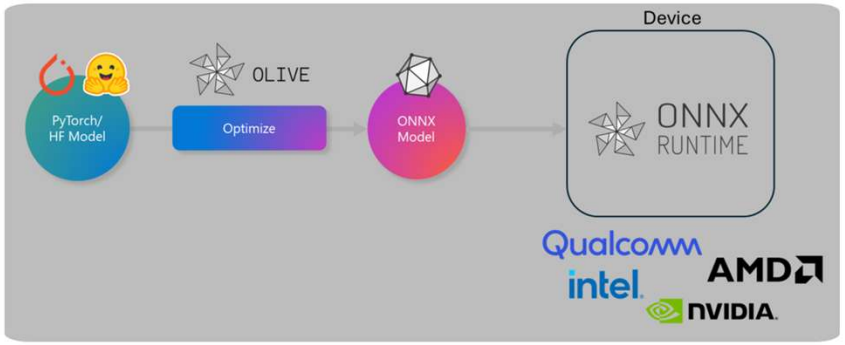
Foundry Local vs Ollama and Llama

	Foundry Local	Ollama	llama.cpp
Primary focus	Application-integrated local AI	Easy local LLM experience	High-performance local inference
Model ecosystem	Foundry Catalog + custom ONNX	Ollama library/custom models	Mostly GGUF
Runtime	ONNX Runtime	Ollama runtime	llama.cpp
Model format	ONNX	Primarily GGUF + supported imports	GGUF
CPU	✓	✓	✓
GPU	✓	✓	✓
NPU	Strong focus	More limited	Limited
Hardware abstraction	Strong	Strong	More developer-controlled
SDK integration	C#, JS, Python, Rust	REST/API/client libraries	Libraries/API
OpenAI-compatible API	✓	✓	Via server
Microsoft ecosystem	Strong	Independent	Independent
Embedded in application	✓	Usually daemon/server	✓

How to run your own models in Foundry Local

- Use Olive to convert and optimize Hugging Face (Safetensor or Pytorch) into ONNX
- Run your optimized model with Foundry Local

Olive to transform models for Foundry Local




```
olive optimize \
  --model_name_or_path Qwen/Qwen2.5-0.5B-Instruct \
  --precision int4 \
  --output_path models/qwen
```

- Olive (**ONNX LIVE**) is a cutting-edge model optimization toolkit with an accompanying CLI that enables you to ship models for the ONNX runtime with quality and performance.
- Quantization: FP32 -> FP16 -> INT8 -> INT4
- Smaller model -> less memory -> faster inference
- Cost: model quality

Src: https://microsoft.github.io/Olive/why-olive.html

Foundry Local vs Cloud

Local	Cloud
Data remains on device	Data sent to service
Can work offline	Requires connectivity
No network latency	Network latency
No per-token inference charge	Usage-based cost
Limited by device hardware	Massive compute available
Smaller models typically	Frontier models available
Hardware varies by user	Controlled infrastructure
Harder fleet/model updating	Centralized deployment
Excellent privacy	Centralized governance

Hybrid approach: Foundry Local + Cloud

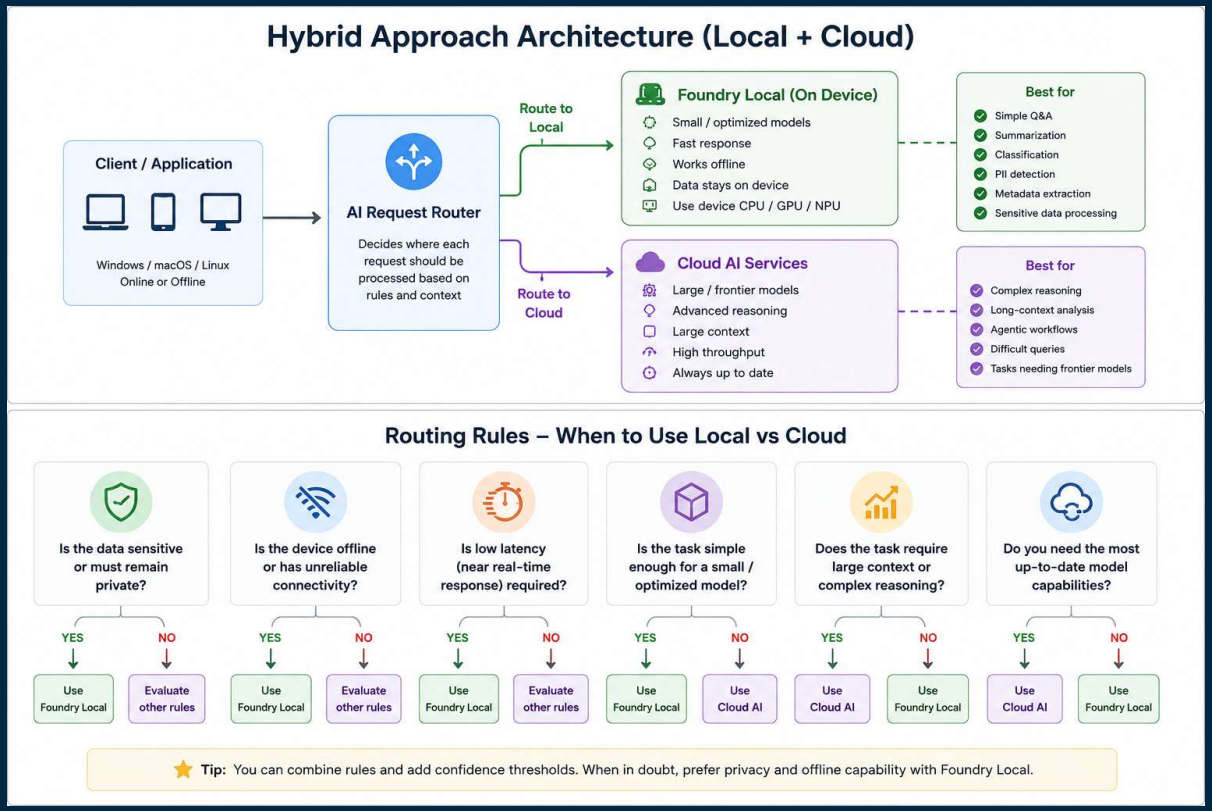
Foundry Local

- Document classification
- Extracting metadata
- Summarization
- Entity extraction
- Sensitive-data detection
- Simple Q&A

Cloud

- complex reasoning
- large-context analysis
- agentic workflows
- difficult queries
- tasks requiring frontier-model capabilities

Hybrid approach architecture



FAQ Foundry Local ?

- Is Foundry local web server + cli tool?
- Why doesn’t it support every model?
- Can I run Foundry Local on a server?
- Does Foundry Local send prompts or output to Microsoft?
- Is Azure subscription needed?

References

- https://learn.microsoft.com/en-us/azure/foundry-local/
- https://github.com/microsoft-foundry/foundry-samples/tree/main



Thank You

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